

Analyse Linux Kernel Code with Sparse

Sparse is a semantic parser written for static analysis of the Linux kernel code. Here's how you can use it to analyse Linux kernel code.



Sparse implements a compiler front-end for the C programming language, and is released under the Open Software License (version 1.1).

You can obtain the latest sources via Git:

```
$ git clone git://git.kernel.org/pub/scm/devel/sparse/sparse.git
```

You can also install it on Fedora using the following command:

```
$ sudo yum install sparse
```

The inclusion of 'C=1' to the `make` command in the Linux kernel will invoke Sparse on the C files to be compiled. Using 'make C=2' will execute Sparse on all the source files. There are a number of options supported by Sparse that provide useful warning and error messages. To disable any warning, use the '-Wno-option' syntax. Consider the following example:

```
void
foo (void)
{
}

int
main (void)
{
    foo();

    return 0;
}
```

Running Sparse on the above `decl.c` file gives the following output:

```
$ sparse -I/usr/include/linux \
-I/usr/lib/gcc/x86_64-redhat-linux/4.7.2/include \
-I/usr/include decl.c
```